
SweetYaar
A little guide for
big grown-ups

Welcome

SweetYaar is a little music-box heart that lives inside a doll. Your child gets two big, friendly buttons. You get the grown-up controls on your phone.

What SweetYaar does



Songs and lullabies

Press a button and the doll plays music from its memory card.



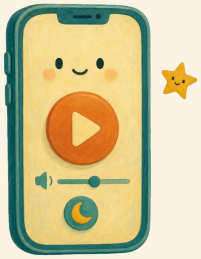
Animal sounds

Press the other button for a moo, a quack or a baa.



A Bluetooth speaker

Play music or a story from your phone, tablet or computer.



In your hands

Choose music, volume and bedtime, or pause the buttons, from the app.



Peace and quiet

Need a breather? Quiet time pauses the doll's buttons for ten minutes, so you get some peace.



Always up to date

New firmware, the doll's built-in software, arrives from the app. No cables.

In this guide

1	Getting started	Charge, switch on, first play
2	Playing with the doll	The two buttons and the little light
3	Using it as a Bluetooth speaker	Pairing your phone
4	The SweetYaar app	Your remote control, settings, adding sounds
5	Keeping the doll up to date	Installing new firmware from the app
6	Bedtime, Quiet time and sleep	Making it fit your routine
7	Care, charging and safety	The battery and how to treat it
8	Troubleshooting	Quick answers
A	Appendix: your own songs and sounds	Files, folders and a script



1 Getting started

Four quick steps and the doll is ready to play.

1. **Charge it.** Plug a USB-C cable into the charging port and use a 5 V USB charger that can supply at least 1.5 A.
2. **Switch it on.** The doll has a small on/off switch. It is meant for storage and travel: normally you leave it on, because the doll sleeps by itself.
3. **Wait for the chime and the green light.** A short “ready” sound plays and the little light blinks green. SweetYaar is ready.
4. **Press the song button.** Music! Press it again for the next song.



The memory card is already inside

SweetYaar ships with a microSD card inside the doll, with songs and animal sounds on it. You can add your own later (see Appendix A at the end).

What is where

Part	What it does
Song button	Plays songs from the current theme (for example lullabies).
Animal button	Plays one animal sound.
Little light	Shows what the doll is doing. See the guide in the next section.
Charging port (USB-C)	For charging. Power only; it is not used to load music or firmware.
On/off switch	Complete power-off for storage and travel.
Memory card slot	Inside the doll.



2 Playing with the doll

The buttons are deliberately simple, so little hands can't get lost.



Song button
music and lullabies

both together = stop



Animal button
animal sounds

Do this	And the doll…
Press the **song** button	plays a song. Press it again to hear the next song.
Press the **animal** button	plays one animal sound. Press it again for another animal.
Press **both** buttons together	stops the sound.
Hold both for **3 seconds**	opens Bluetooth pairing for 60 seconds (see section 3).
Keep holding both to **10 seconds**	forgets every paired phone (see section 3).

A song stops when it ends. Turn on **Loop songs** in the app and the next song starts by itself. Fancy some surprises? Each theme also has a **shuffle** switch in the app's Settings, which plays its songs in random order.

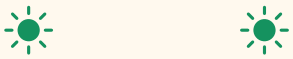
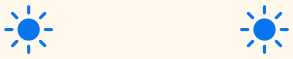


Sleeping and waking

After about 10 minutes with nothing happening, the doll falls asleep to protect the battery. Pick it up or move it and it wakes with its ready chime. Music that was playing does not resume, and any Bluetooth connection has to be made again. You can change these times in the app.

What the little light means

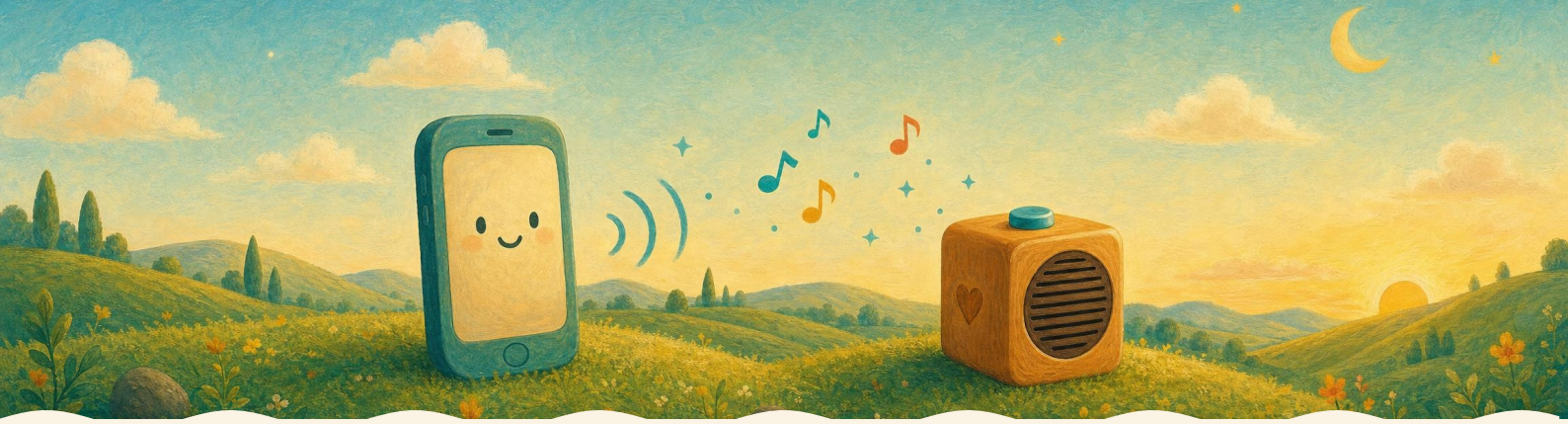
A single light shows the doll's state at a glance. In the pictures below, a dot with rays is one blink, and the closer together the blinks are, the faster the light is flashing.

	Solid yellow	Starting up.
	**Green, slow blink**	Ready. Press a button.
	**Green, quick blink**	Playing a song or animal sound.
	**Blue, slow blink**	A phone is connected as a speaker, nothing playing.
	**Blue, quick blink**	Music is streaming from your phone.
	**Blue, rapid blink**	Pairing is open for 60 seconds.
	**Purple**	Quiet time is on: the buttons are paused. (Lit most of the time, with a brief flicker.)
	**Red, one flash**	All paired phones were forgotten.
	**Red, rapid blink**	Something needs attention (for example the memory card is missing or can't be read).
	Yellow, quick blink	Installing an update. Keep the app open.
	**Off**	Sleeping.

Sounds the doll makes

Besides music, the doll plays a few short chimes of its own. They work even without a memory card, and they use the doll's current volume, so they are softer at bedtime.

Chime	When you hear it
Ready	Once when the doll has started and is ready to play.
Pairing	Once when you open pairing (hold both buttons for 3 seconds).
Charger plugged in	A rising three-note "sunrise", once, when you plug in the charger and charging really starts.
Low battery	Three soft notes when the battery drops to "charge soon" or "charge now" while the doll is not charging.
Error	Something needs attention, for example the memory card is missing.



3 Using it as a Bluetooth speaker

SweetYaar can play anything from your phone, tablet or computer, like a story, a podcast or your own playlist.

Connect your phone for the first time

1. Hold **both buttons for 3 seconds**. A chime plays once and the light blinks blue rapidly. You now have **60 seconds**.
2. On your phone, open **Bluetooth settings** and choose **SweetYaar** from the list.
3. If your phone asks, tap **Pair**. There is no code to type.
4. Play something on your phone. The sound comes out of the doll.

- **You only pair once.** Next time, you don't need to open pairing again: with the doll awake and nearby and your phone's Bluetooth on, the phone connects by itself.
- While the phone is connected, the doll's buttons pause and the app shows "Used as a speaker". **Use the phone's own volume control.**
- Only one phone can stream at a time. Disconnect the first phone before connecting a second.
- **To pair the app, pause any music first.** While a phone is streaming music to the doll, the doll refuses to pair a new app, even in the 60-second window. Pause the music, then connect (see section 4).
- When the phone disconnects, the doll goes back to its songs and animal sounds.
- **The speaker and the SweetYaar app are paired separately:** pair each one while pairing is open. Phones stream best; a computer can have small gaps.

Forgetting all paired phones

Hold both buttons for **10 seconds**. The light flashes red once and every paired phone and app is removed. Use this if you give the doll away or want a fresh start. Afterwards, also choose "Forget this device" for SweetYaar in your phone's Bluetooth settings, then pair again.



4 The SweetYaar app

Your remote control: pick the music, set the volume, set bedtime.

Before you start

- The app is a website, so there is nothing to download from an app store. Open **sweetyaar.zachmoshe.com** in **Chrome** on an Android phone, or in Chrome or Edge on a computer. (Safari and other iPhone browsers don't support the Bluetooth feature the app needs.)
- You can add it to your home screen to use it like a normal app. It also opens without internet. Talking to the doll uses Bluetooth only: no accounts, no cloud.

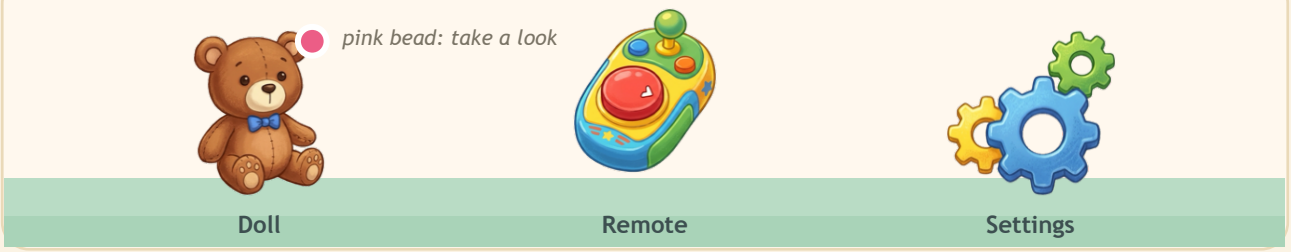
Connect for the first time

1. Make sure the doll is awake and nearby (and not streaming music from a phone), then hold **both buttons for 3 seconds** to open pairing.
2. In the app, tap **Connect to SweetYaar** and choose the doll from the list your browser shows.
3. Accept the Bluetooth pairing request on your phone.

Afterwards, just open the app and tap **Connect to SweetYaar** (wake the doll first if it sleeps). A phone the doll doesn't know yet, or one you forgot with the 10-second hold, gets **"Device not approved"**: hold both buttons for 3 seconds and tap Connect again.

Finding your way around

Once connected, three little toys standing on the grass at the bottom of the screen are the app's menu. Tap one to switch page.



Page	What it is for
Doll (teddy)	How the doll is doing: battery and charging, its firmware version, and updates. A **pink bead** on the teddy's ear means it is worth a look: an update you haven't seen, a low battery, or a charger problem.
Remote	The remote control: play, stop, volume, theme, Quiet time. While a phone streams music it says "Used as a speaker".
Settings (gears)	Everything permanent: name, defaults, bedtime, sleep, themes and songs.

The remote screen

Control	What it does
Play song	Plays a song from the current theme. Tap again for the next one.
Animal	Plays an animal sound. Tap again for another.
Stop	Stops whatever is playing.
Loop songs	Keeps playing one song after another. It switches itself off after Stop, an animal sound, Quiet time, phone streaming or a restart.
Volume	Sets the doll's volume for its own songs and sounds. (It does not change the volume of phone streaming.)
Theme	Chooses which group of songs the song button plays. Lasts until the doll sleeps; set a permanent default in Settings.
Quiet time	Stops the sound and makes the doll ignore its buttons for 10 minutes, so you get some quiet (for example during a phone call). The app shows the time left. Switch it off any time. You can still change the volume; it applies when Quiet time ends.
Daytime / Bedtime	Shows the current mode and lets you switch it for now.
Battery	Shows the battery status, and when the doll is charging. Tap it to open the Doll page.

Settings and your own sounds

Settings

Change as many things as you like, then tap **Save** once.

Section	What you can set
General	The doll's Bluetooth name, the volume it starts at, and the theme it starts with.
Bedtime mode	Turn it on or off, choose the bedtime theme, the start and end time, and a volume limit.

Section	What you can set
Sleep	Turn automatic sleep on or off and how long the doll waits before it sleeps.
Themes	A **theme** is a group of songs, such as lullabies or nature sounds. Switch a whole theme off and the doll skips it. Shuffle plays a theme's songs in random order.
Songs and animals	See every song and animal sound, and switch individual ones off by tapping a row. Anything switched off is skipped, while the rest of its theme keeps playing. A file the doll can't play is skipped and shown with the reason.



Volume and theme: "now" versus "at start"

The volume and theme on the Remote page are for right now. The ones in Settings are what the doll uses **each time it wakes up**. Saving Settings does not change what is playing right now.

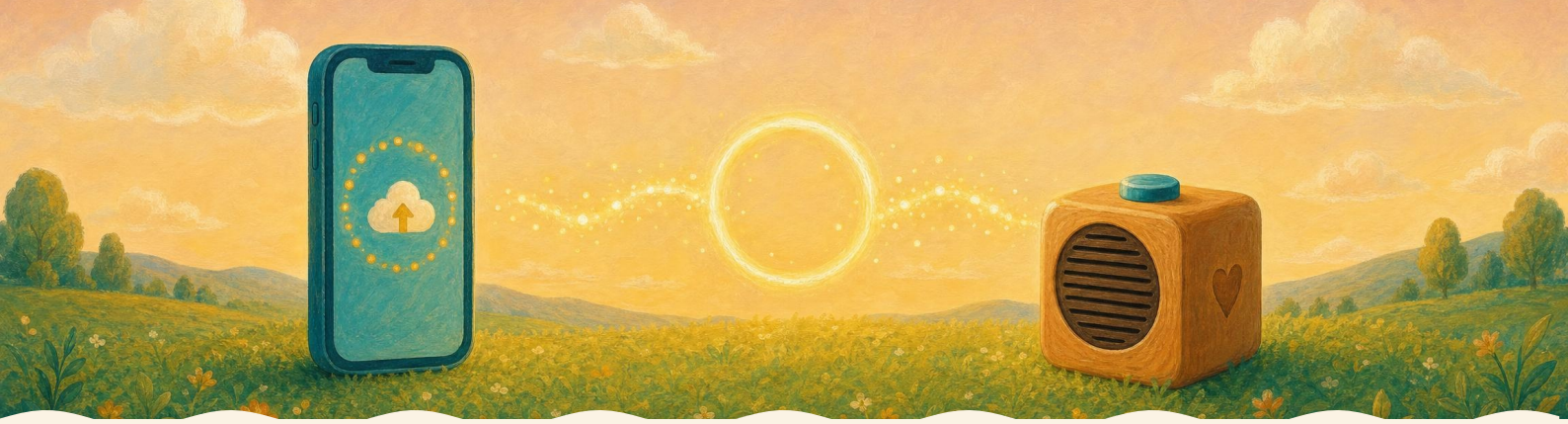
The Doll page

The Doll page shows how the doll is doing. At the top is the charge status (for example charging, finished, or no external power). Below it, charger notes appear only when something needs a look. *About this doll* shows the firmware version and whether a newer one is available (see section 5). When the battery is getting low, a sleepy teddy appears as a reminder.



Adding your own songs and sounds

The doll's songs and animal sounds live on the microSD card inside it. The app can organise them, but it can't add files: to add your own, you copy files onto the card from a computer. You can add **songs**, whole new **themes**, or even **record your own voice** for the animal prompts. This needs some basic terminal skills, such as running a script, and it is all explained in **Appendix A** at the end.



5 Keeping the doll up to date

New firmware, the doll's built-in software, is installed from the app over Bluetooth. No cable, no computer, no Wi-Fi.

How you find out

When you connect, the app checks whether a newer version has been published. If there is one, a **pink bead** appears on the Doll page. The bead goes away once you have opened that page on that phone. A very old doll that the app can no longer control shows an **Update needed** card instead, and its remote stays locked until it is updated.

Updating

1. Open the **Doll** page. *About this doll* shows the doll's version and the new one, with what's new. Tap **Update to ….**
2. Read **Before you start**: keep the screen open, keep the doll near you, and expect no songs for about a minute. The doll needs enough battery for the update, or to be charging. If the battery is too low, the update won't start until you connect the charger.
3. Tap **Start update**. The app downloads the update and checks it, restarts the doll into update mode (the light blinks yellow), reconnects by itself with no pairing, and sends the update with a progress ring and the time left.
4. The doll installs it and restarts. The app confirms **SweetYaar is up to date** and shows what's new, then returns to the remote.



Good to know

If the connection drops (for example the phone locks): the app shows "Update paused" and keeps reconnecting, then carries on where it stopped. The doll waits 2 minutes and then goes back to normal on its current version.
If you switch the doll off mid-update: it comes back on its current version, and the app says "Update didn't finish". Tap **Try again**; nothing is damaged.
To stop: tap **Cancel update** and confirm. The doll discards the partial update and restarts as before.
You can also **Reinstall** the current version from the Doll page to repair a doll.

If a new version turns out to have a problem while starting, the doll automatically goes back to the previous version.



6 Bedtime, Quiet time and sleep

Three gentle helpers that make SweetYaar fit your routine.



Bedtime mode

Every day between a start and an end time you choose (the factory setting is 18:30 to 06:30), the doll automatically switches to a calmer mode:

- It plays songs from your **bedtime theme** (for example lullabies).
- It limits the volume to the **bedtime volume limit** you set (factory setting 65%). This applies to songs and animal sounds. Your normal volume setting is kept for daytime.
- It does not affect music streamed from a phone, which is controlled by the phone.

The doll does not know the time by itself. Every time you connect the app, it tells the doll the correct time. Until then the app shows “clock not set” and Bedtime mode stays off. You can switch between Daytime and Bedtime for now from the Remote page; the schedule takes over again at the next boundary.

Quiet time

Want the doll to stop, and stay quiet, for a while, say during a phone call or when you need five minutes of peace? Switch on **Quiet time** in the app. The sound stops and the doll ignores its buttons for **10 minutes**, or until you switch it off. The light turns purple. A phone that is already streaming is not stopped.

Automatic sleep

After this much idle time…	…the doll sleeps (factory setting)
Nothing has been touched	10 minutes
It was woken by movement, but nobody pressed a button	2 minutes
The app is connected but you are not doing anything	2 minutes

Sleep never starts while music plays. The doll wakes when moved or plugged in, and starts fresh, so reconnect your phone or the app if you need them.



7 Care, charging and safety

The battery

Type	Removable lithium-ion 18650 cell with built-in protection (button top), 3.6–3.7 V nominal and 4.2 V when full.
Capacity	2400 mAh or 3400 mAh, depending on the model.
Charging	Through the USB-C port, at up to 1 A. Use a 5 V charger that provides at least 1.5 A.
Temperature	Charging only runs while the battery is between 0 °C and 45 °C, and the doll stops charging by itself outside that range. A six-hour safety timer also limits a charge.

Charging

- When charging really starts, the doll plays a short rising “sunrise” chime once. No chime usually means the charger isn’t giving power.
- The doll can play while charging, but it charges slower then.
- If it doesn’t charge, check the cable and the charger (a 5 V charger that can supply at least 1.5 A) and try again.
- Charge it where you can see it, on a hard surface, and unplug it once it is full if you are going away.
- Switch the doll off with its on/off switch if you will not use it for many weeks.

Safety

- **Adults only inside.** The battery, electronics and memory card are inside the doll. Only an adult should open it, and never a child.
- Use only a protected 18650 cell of the same type. An unprotected or damaged cell can overheat.
- Stop using the doll and take the battery out if it is swollen, dented, leaking, smells odd or gets hot while charging.
- Never crush, puncture, heat, short-circuit or wet the battery, and don’t charge a damaged cell.
- Keep the doll away from heat, direct sun, a radiator or a hot car.
- Don’t throw used batteries in the bin. Take them to a battery collection point.
- Do not use a damaged cable. Do not wash or soak the doll’s electronics.
- Check the doll regularly for loose parts and wear, as with any toy.



8 Troubleshooting

What's happening	Try this
The doll doesn't react to the buttons.	It may be asleep, so move it gently and wait for the chime. Check that the on/off switch is on and that it is charged. Is the light purple? Quiet time is on; switch it off in the app.
The light blinks red rapidly (and there was an error chime).	Something is wrong, most often the memory card: it is missing, loose or can't be read. Switch the doll off, check that the card is seated and holds the SweetYaar folders, then switch on again.
The light is green but no songs play.	The current theme may have no songs, or its songs and the theme are switched off in Settings. Pick another theme in the app. Also check that Quiet time is off.
My phone can't find SweetYaar.	Hold both buttons for 3 seconds first (pairing only opens for 60 seconds), and keep the phone close.
The phone connected but there's no sound.	Check the phone's volume and that SweetYaar is chosen as the audio output. Then try disconnecting and reconnecting.
The app says "Device not approved".	The doll doesn't know this phone yet. Hold both buttons for 3 seconds and tap Connect again within the minute.
The app asks me to pause the music and pair again.	While a phone streams music to the doll, a new app can't pair. Pause the music, hold both buttons for 3 seconds, and connect again.
The app says "Update needed".	The doll's firmware is too old for this app. Open the Doll page and update it (keep the doll near you and the screen open).
The app says "Update didn't finish".	Tap Try again. Make sure the battery isn't low, keep the doll close and don't switch it off during the update.
The app says "Someone else took control".	Another approved phone connected. Tap Connect to take control back.
The app can't connect.	Move the doll to wake it, use Chrome (or Edge on a computer), and make sure Bluetooth is on. If it still fails, forget SweetYaar in your phone's Bluetooth list and pair again.
The sound crackles.	Make sure the battery isn't low and try a lower volume. If it persists after charging, contact support.
Bedtime doesn't start.	Open the app once so the doll learns the time (it shows "clock not set" until then). Check that Bedtime mode is on in Settings.
I want to start over.	Hold both buttons for 10 seconds to forget every paired phone, then pair again.


Technical stuff
Appendix A



A Adding your own content

You need a computer and a card reader. Nothing here is needed for everyday use.

Before you start

- **Switch the doll off** with its on/off switch before taking the microSD card out, and switch it on again after the card is back in.
- The card must be **FAT32**, like the one that came with the doll. Keep the folders exactly as shown below.
- Your computer may leave hidden files on the card, such as `.DS_Store` or `._cat.wav`. You don't need them, so delete them if you like. If you leave them, the doll simply ignores them; it only costs a little extra time when it scans the card.
- The doll reads the card when it starts, so new songs and sounds appear after a restart (switch it off and on, or let it sleep and wake).

What's on the card

```
/songs/
    lullabies/              one folder per theme
        metadata.json       the theme's settings (name, shuffle...)
        01 Twinkle Twinkle.wav
        02 Rock a Bye Baby.wav
    nature-sounds/
        ...
/animals/
    metadata.json
    dog.wav                 each animal file: a question, then the animal
    cat.wav
    ...
/config.json                doll settings (the app edits this; leave it alone)
```

A.1 Adding songs

1. Make a **new folder** inside `/songs`. Its name becomes a new theme in the app.
2. Optionally, give the theme a name and settings with a `metadata.json` file in that folder (an example is under "Theme settings" below). Copying one from an existing theme and changing its *name* works just as well. Without one, the theme is named after its folder. Emojis are welcome in the theme's name.

3. Copy your songs into the folder, as WAV files in the format below. Songs play in file-name order; to choose the order, put a track number in front of each name (01, 02, …). The number only sets the order, and the app doesn't show it.
4. Put the card back, switch the doll on, and open **Settings** in the app: your new theme and songs are there, and you can switch any of them off.

Song file	Requirements
Format	**WAV**, uncompressed PCM
Sample rate and depth	**44.1 kHz, 16-bit**
Channels	**Mono** (recommended: half the space, same sound from one speaker). Stereo also plays.

To turn an MP3 or other recording into the right kind of WAV, for example with the free ffmpeg tool (a download link is on its site):

```
ffmpeg -i input.mp3 -ar 44100 -ac 1 -c:a pcm_s16le output.wav
```

Names. The app shows each song's file name, tidied up: no ".wav", no leading track number such as "01 -", and underscores become spaces. Right-to-left languages (for example Hebrew and Arabic) and emojis work in file names and theme names.

Theme settings. Each theme folder has a small *metadata.json* file. *name* is the theme's name in the app, *shuffle* plays its songs in random order, and *disabledSongs* lists songs that are switched off (the app fills this in for you). You can add an optional *attribution* to credit where the sounds come from; the app shows it under the current theme. It is written in **Markdown**, so links written as `[label](https://…)` become tappable. Keep it to one short line.

```
{
  "schemaVersion": 2,
  "name": "Lullabies",
  "shuffle": false,
  "disabledSongs": [],
  "attribution": "[Pixabay](https://pixabay.com/sound-effects/)"
}
```

A.2 Animal sounds

Each animal file on the card is a short little scene: **the doll asks a question** ("What does the dog say?"), pauses, and then you hear the **real animal**. The question and the animal sound are kept as two separate recordings and are joined by a small script. That means:

- The **animal sounds** are real recordings that come with the project, and stay the same in every language.
- The **questions** are separate. They can be in another language, in another voice, or **in your own voice**. Record one short question per animal, run the script once, and copy the result to the card.

The 13 animals: dog, cat, sheep, cow, rooster, elephant, fish, duck, horse, lion, monkey, frog, owl.

Recording your own questions

1. Make a folder for your recordings, for example `my_questions`.
2. Record one short question for each animal, for example “What does the dog say?”, about one to two seconds each, in a quiet room. Phone voice memos are fine; convert them to WAV (mono is best, any sample rate is accepted).
3. Name each file after its animal: `dog.wav`, `cat.wav`, `sheep.wav` … one file for each of the 13 animals. Leave out an animal and it is simply left out.
4. Run the script (below). It trims the silence around your voice, evens out the volume, adds a pause and joins your question to the animal sound.
5. Copy the results to the card’s `/animals` folder, replacing the files there.

Running the script

The script is part of the SweetYaar project on GitHub: github.com/zachmoshe/sweetyaar. Download it, open a terminal in the project folder and, the first time only, set up its tools:

```
make setup
```

The command is `make animals`, and it takes three options. You only need the ones you want to change:

Option	What it does
`PROFILE=`	Which ready-made set to build: it picks the spoken questions and which animals. The default is `default`: all 13 animals with the ready-made English questions. Also available: `en-tts` (English) and `he-tts` (Hebrew).
`TTS=`	A folder with **your own** question recordings. They replace the profile’s ready-made questions (the profile still decides which animals are built). Leave it out to use the ready-made ones. (The name comes from “text-to-speech”, how the ready-made questions were made.)
`OUT=`	The folder the finished files are written to. It is created if it doesn’t exist, and nothing already in it is deleted. Leave it out and they go to a scratch folder inside the project.

The simplest command: ready-made English questions, all 13 animals, written to a folder on your Desktop:

```
make animals OUT=~/Desktop/animals
```

“I recorded my own questions”: use your recordings instead of the ready-made ones. You don’t need `PROFILE=`; the default builds all 13 animals and your recordings replace its questions:

```
make animals TTS=~/Documents/my_questions OUT=~/Desktop/animals
```

The ready-made Hebrew questions instead of English:

```
make animals PROFILE=he-tts OUT=~/Desktop/animals
```

Everything else you can change is added to the same command:

Want to…	Add this
A longer or shorter pause after the question	`PAUSE=1.0` (seconds; the default is 0.7)
Build only some animals	`ANIMALS="dog cat"`

The script prints one line per animal and ends with `Done: 13/13 built.` If a recording is missing it says so and ends with `!! INCOMPLETE` and the animals it left out. Copy **all** the files in the output folder, including `metadata.json`, to the card's `/animals` folder.